Regression

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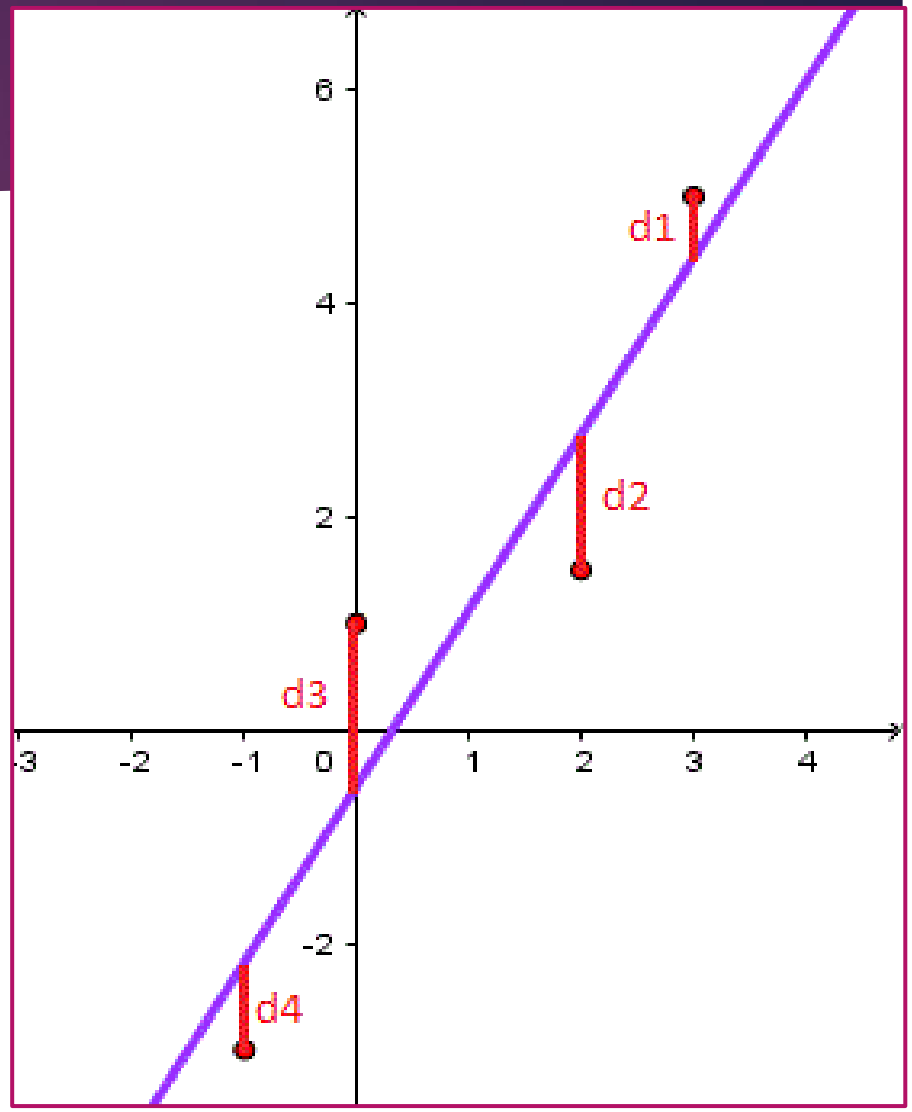
Introduction

- Linear regression and modelling problems are presented along with their solutions. Also a linear regression calculator and grapher may be used to check answers and create more opportunities for practice.
- If the plot of n pairs of data (x , y) for an experiment appear to indicate a "linear relationship" between y and x, then the method of least squares may be used to write a linear relationship between x and y.
- The least squares regression line is the line that minimizes the sum of the squares (d1 + d2 + d3 + d4) of the vertical deviation from each data point to the line (see figure below as an example of 4 points).



- The least square regression line for the set of n data points is given by the equation of a line in slope intercept form:
- $y = a x + b$

$$a = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - (\sum_{i=1}^{n} x_i)^2}$$

$$b = \frac{1}{n}(\sum_{i=1}^{n} y_i - a\sum_{i=1}^{n} x_i)$$

Troubleshooting --Problem 1

Consider the following set of points: {(-2 , -1) , (1 , 1) , (3 , 2)}

a) Find the least square regression line for the given data points.

b) Plot the given points and the regression line in the same rectangular system of axes.

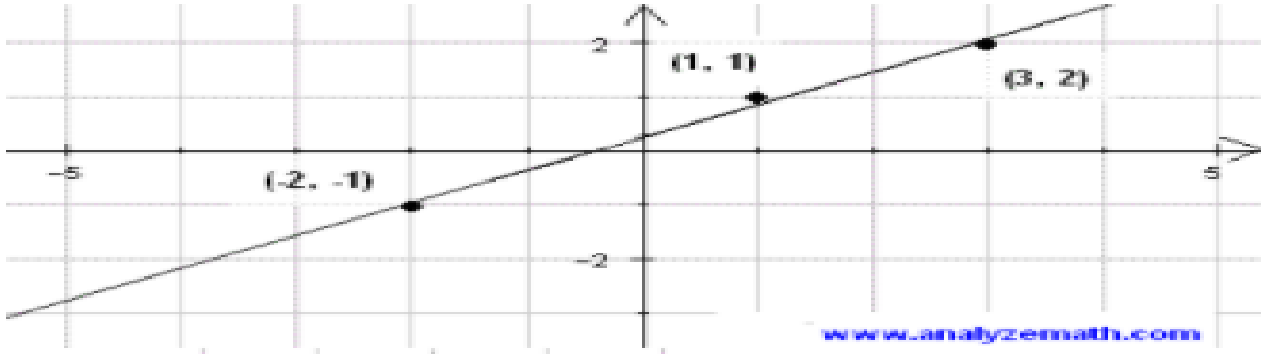
x	y	x y	x^2
-2	-1	2	4
1	1	1	1
3	2	6	9
$\Sigma x = 2$	$\Sigma y = 2$	$\Sigma xy = 9$	$\Sigma x^2 = 14$

We now use the above formula to calculate a and b as follows

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (3*9 - 2*2) / (3*14 - 2^2) = 23/38$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/3)(2 - (23/38)*2) = 5/19$

b) We now graph the regression line given by y = a x + b and the given points.



Problem 2

a) Find the least square regression line for the following set of data

{(-1 , 0),(0 , 2),(1 , 4),(2 , 5)}

b) Plot the given points and the regression line in the same rectangular system of axes.

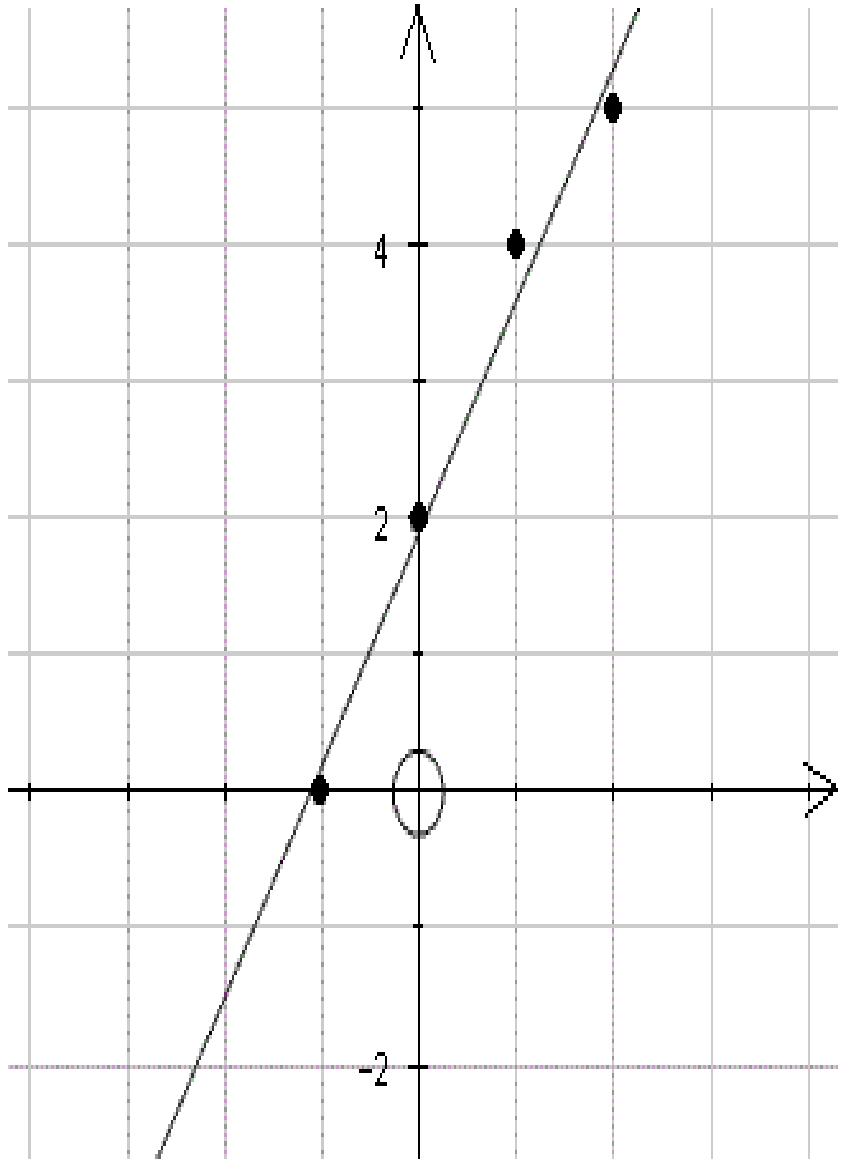
x	y	x y	x^2
-1	0	0	1
0	2	0	0
1	4	4	1
2	5	10	4
$\Sigma x = 2$	$\Sigma y = 11$	$\Sigma x\, y = 14$	$\Sigma x^2 = 6$

We now use the above formula to calculate a and b as follows

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (4*14 - 2*11) / (4*6 - 2^2) = 17/10 = 1.7$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/4)(11 - 1.7*2) = 1.9$

b) We now graph the regression line given by y = ax + b and the given points.



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Problem 3

- The values of y and their corresponding values of y are shown in the table below

X	0	1	2	3	4
y	2	3	5	4	6

a) Find the least square regression line $y = a x + b$.

b) Estimate the value of y when $x = 10$.

x	y	x y	x^2
0	2	0	0
1	3	3	1
2	5	10	4
3	4	12	9
4	6	24	16
$\Sigma x = 10$	$\Sigma y = 20$	$\Sigma x\, y = 49$	$\Sigma x^2 = 30$

We now calculate a and b using the least square regression formulas for a and b.

$a = (n\Sigma x\, y - \Sigma x \Sigma y) / (n\Sigma x^2 - (\Sigma x)^2) = (5*49 - 10*20) / (5*30 - 10^2) = 0.9$

$b = (1/n)(\Sigma y - a\, \Sigma x) = (1/5)(20 - 0.9*10) = 2.2$

b) Now that we have the least square regression line $y = 0.9\, x + 2.2$, substitute x by 10 to find the value of the corresponding y.

$y = 0.9 * 10 + 2.2 = 11.2$

Problem 4

- The sales of a company (in million dollars) for each year are shown in the table below.

x (year)	2005	2006	2007	2008	2009
y (sales)	12	19	29	37	45

- a) Find the least square regression line $y = a x + b$.
- b) Use the least squares regression line as a model to estimate the sales of the company in 2012.

We first change the variable x into t such that t = x - 2005 and therefore t represents the number of years after 2005. Using t instead of x makes the numbers smaller and therefore manageable. The table of values becomes.

t (years after 2005)	0	1	2	3	4
y (sales)	12	19	29	37	45

We now use the table to calculate a and b included in the least regression line formula.

t	y	t y	t^2
0	12	0	0
1	19	19	1
2	29	58	4
3	37	111	9
4	45	180	16
$\Sigma x = 10$	$\Sigma y = 142$	$\Sigma xy = 368$	$\Sigma x^2 = 30$

We now calculate a and b using the least square regression formulas for a and b.

$a = (n\Sigma t\, y - \Sigma t \Sigma y) / (n\Sigma t^2 - (\Sigma t)^2) = (5*368 - 10*142) / (5*30 - 10^2) = 8.4$

$b = (1/n)(\Sigma y - a\ \Sigma x) = (1/5)(142 - 8.4*10) = 11.6$

b) In 2012, t = 2012 - 2005 = 7

The estimated sales in 2012 are: y = 8.4 * 7 + 11.6 = 70.4 million dollars.

It will be Continued….